ME551 Vibrations in Linear Systems

Lecture 13

Multiple Degree of Freedom Systems 5

(Nonconservative Syst.)

Chap 4. Multi-Degree-of-Freedom System

4.1 Equations of Motion
4.2 Geometric Representation of Motion. Equilibrium Points
4.3 Stability of Equilibrium Points. Liapunov Direct Method
4.4 Linearization about Equilibrium Points
4.5 Stability of Linearized Systems
4.6 Linear Conservative Natural Systems. Symmetric Eigenvalue Problem
4.7 Gyroscopic Conservative Systems
4.8 Nonconservative Systems. Nonsymmetric Eigenvalue Problem
4.9 Response of Discrete Systems to Harmonic Excitations
4.10 Response of Multi-DOF Systems to Arbitrary Excitations
i. Undamped natural systems
ii. Systems with proportional viscous damping
iii. Systems with small damping
iv. Systems with structural damping
v. Undamped gyroscopic systems
vi. General nonconservative systems
4.11 Discrete-Time Systems

Linearized system in general:

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \left(\underset{\text{sym}}{\boldsymbol{C}^*} + \underset{\text{skew sym}}{\boldsymbol{G}}\right)\dot{\boldsymbol{q}} + \left(\underset{\text{sym}}{\boldsymbol{K}} + \underset{\text{skew sym}}{\boldsymbol{H}}\right)\boldsymbol{q}(t) = 0$$

$$\downarrow$$

State Space Representation: $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}(t),\ \boldsymbol{x}(t) = \begin{Bmatrix} \boldsymbol{q}(t) \\ \dot{\boldsymbol{q}}(t) \end{Bmatrix}$

$$\downarrow \Leftarrow \boldsymbol{x}(t) = \boldsymbol{x}e^{\lambda t}, \lambda = \sigma + i\omega$$

Eigenvalue Problem:

$$\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x},\ \boldsymbol{A} \neq \boldsymbol{A}^T$$

Formulation of state equation and eigenvalue problem for an un-symmetric matrix $\boldsymbol{A}$

$\boldsymbol{M}\ddot{\boldsymbol{q}}+\left(\boldsymbol{C}^{*}+\boldsymbol{G}\right)\dot{\boldsymbol{q}}+\left(\boldsymbol{K}+\boldsymbol{H}\right)\boldsymbol{q}(t)=0$, where $\boldsymbol{M},\boldsymbol{C}^{*}$ & $\boldsymbol{K}$: real sym. PD and $\boldsymbol{G}$ & $\boldsymbol{H}$: real skew sym.

$\downarrow$

$$\ddot{\boldsymbol{q}}=-\boldsymbol{M}^{-1}\left(\boldsymbol{C}^{*}+\boldsymbol{G}\right)\dot{\boldsymbol{q}}-\boldsymbol{M}^{-1}\left(\boldsymbol{K}+\boldsymbol{H}\right)\boldsymbol{q}$$

$\downarrow\Leftarrow$ Augmentation by $\dot{\boldsymbol{q}}=\dot{\boldsymbol{q}}$

$$\underbrace{\frac{d}{dt}\begin{Bmatrix}\boldsymbol{q}\\ \dot{\boldsymbol{q}}\end{Bmatrix}}_{\dot{\boldsymbol{x}}}=\underbrace{\begin{bmatrix}0 & \boldsymbol{I}\\ -\boldsymbol{M}^{-1}\left(\boldsymbol{K}+\boldsymbol{H}\right) & -\boldsymbol{M}^{-1}\left(\boldsymbol{C}^{*}+\boldsymbol{G}\right)\end{bmatrix}}_{\equiv\boldsymbol{A}:\ (2n\times 2n)\text{ real unsymmetric}}\underbrace{\begin{Bmatrix}\boldsymbol{q}(t)\\ \dot{\boldsymbol{q}}(t)\end{Bmatrix}}_{\equiv\boldsymbol{x}(t)}:$$

$$\dot{\boldsymbol{x}}=\boldsymbol{A}\boldsymbol{x}(t)$$

$\downarrow\Leftarrow$ $\boldsymbol{x}(t)=\boldsymbol{x}e^{\lambda t},\lambda=\sigma+i\omega$ & $\boldsymbol{x}$: complex

$\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{x}$: Algebraic eigenvalue problem

$\downarrow\Leftarrow$ Computational techniques in Chap. 6

$\lambda_j,\boldsymbol{x}_j,j=1,2,\ldots,2n$

$\boldsymbol{x}_i^T\boldsymbol{A}\boldsymbol{x}_j\neq 0,\boldsymbol{x}_i^T\boldsymbol{x}_j\neq 0$: Orthogonality not satisfied by $\boldsymbol{x}_j$'s!

since $\boldsymbol{A}$ is unsymmetric.

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• Adjoint Eigenvalue Problem (EVP)

$$C.E.: \det\left[\boldsymbol{A}-\lambda\boldsymbol{I}\right]=\det\left[\left(\boldsymbol{A}-\lambda\boldsymbol{I}\right)^T\right]=\det\left[\boldsymbol{A}^T-\lambda\boldsymbol{I}\right]=0$$

$\therefore$ Eigenvalues for $\boldsymbol{A}$ and $\boldsymbol{A}^T$ are the same!

$\downarrow$

$$\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{x}:\text{EVP} \quad \Leftrightarrow \quad \boldsymbol{A}^T\boldsymbol{y}=\lambda\boldsymbol{y}:\ \text{Adjoint EVP!}$$

$\downarrow \Leftarrow$ *transpose*

$$\boldsymbol{A}\boldsymbol{x}_i=\lambda_i\boldsymbol{x}_i, i=1,2,\ldots,2n \quad \Leftrightarrow \quad \boldsymbol{y}_j^T\boldsymbol{A}=\lambda_j\boldsymbol{y}_j^T, j=1,2,\ldots,2n$$

$\boldsymbol{x}_i$: Right eigenvector $\qquad$ $\boldsymbol{y}_j$: Left eigenvector

$\downarrow \Leftarrow$ pre-multiply by $\boldsymbol{y}_j^T$ $\qquad$ $\downarrow \Leftarrow$ post-multiply by $\boldsymbol{x}_i$

$$\underbrace{\boldsymbol{y}_j^T\boldsymbol{A}\boldsymbol{x}_i=\lambda_i\boldsymbol{y}_j^T\boldsymbol{x}_i \qquad \boldsymbol{y}_j^T\boldsymbol{A}\boldsymbol{x}_i=\lambda_j\boldsymbol{y}_j^T\boldsymbol{x}_i}_{\text{subtraction}}$$

$$\left(\lambda_i-\lambda_j\right)\boldsymbol{y}_j^T\boldsymbol{x}_i=0$$

$\downarrow \Leftarrow$ When $\lambda_i \neq \lambda_j$ & eigenvector normalization

$$\boldsymbol{y}_j^T\boldsymbol{x}_i=\delta_{ij};\ \boldsymbol{y}_j^T\boldsymbol{A}\boldsymbol{x}_i=\lambda_j\delta_{ij}:\ \text{Bi-orthonormality!}$$

Summary of biorthogonality:

Right Eigenvectors

$$\boldsymbol{X}=\left[\boldsymbol{x}_1\boldsymbol{x}_2\ldots\boldsymbol{x}_{2n}\right] \text{ from } \overset{\text{EVP}}{\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{x}}$$

Bi-orthogonality

$$\boldsymbol{y}_j^T\boldsymbol{x}_i=\delta_{ij}; \quad \boldsymbol{y}_j^T\boldsymbol{A}\boldsymbol{x}_i=\lambda_j\delta_{ij}$$

Left Eigenvectors $\qquad$ Adjoint EVP

$$\boldsymbol{Y}=\left[\boldsymbol{y}_1\boldsymbol{y}_2\ldots\boldsymbol{y}_{2n}\right] \text{ from } \boldsymbol{y}^T\boldsymbol{A}=\lambda\boldsymbol{y}^T$$

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- **Orthogonality in terms of Eigen Matrices**

$$\boldsymbol{X} = \left[\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{2n}\right] \text{ from } \boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x}\text{: Right eigenmatrix of } \boldsymbol{A}$$

$$\boldsymbol{Y} = \left[\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_{2n}\right] \text{ from } \boldsymbol{A}^T \boldsymbol{y} = \lambda \boldsymbol{y}\text{: Left eigenmatrix}$$

$$\downarrow \Leftarrow \boldsymbol{y}_i^T \boldsymbol{x}_j = \delta_{ij} \left(\text{Bi-orthonormalization}\right); \boldsymbol{y}_i^T \boldsymbol{A}\boldsymbol{x}_j = \lambda_i \delta_{ij}$$

$$\boldsymbol{Y}^T \boldsymbol{X} = \boldsymbol{I}$$

$$\boldsymbol{Y}^T \boldsymbol{A}\boldsymbol{X} = \boldsymbol{\Lambda}$$

$$\downarrow \Leftarrow \boldsymbol{Y}^T = \boldsymbol{X}^{-1} \Leftarrow \boldsymbol{Y}^T \boldsymbol{X} = \boldsymbol{I}$$

$$\boldsymbol{X}^{-1}\boldsymbol{A}\boldsymbol{X} = \boldsymbol{\Lambda} \left(\text{A similarity transformation! } \text{See page. 9}\right)$$

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- **Dual Expansion Theorem Using Bi-orthonormality**

$$\boldsymbol{v} = \sum_{i=1}^{2n} a_i \boldsymbol{x}_i \in \mathbb{R}^{2n\times 1} \quad \Leftrightarrow \quad \boldsymbol{v} = \sum_{i=1}^{2n} b_i \boldsymbol{y}_i \in \mathbb{R}^{2n\times 1}$$

$\downarrow \Leftarrow$ pre-multiply by $\boldsymbol{y}_j^T$ $\qquad$ $\downarrow \Leftarrow$ pre-multiply by $\boldsymbol{x}_j^T$

$$\underbrace{\boldsymbol{y}_j^T \boldsymbol{v} = a_j \qquad\qquad \boldsymbol{x}_j^T \boldsymbol{v} = \boldsymbol{b}_j}$$

Expansion of a vector in terms of right(left) eigenvectors for an unsymmetric matrix $\boldsymbol{A}$ $\left(\boldsymbol{A}^T\right)$ can be done easily with left(right) eigenvectors for the transposed matrix $\boldsymbol{A}^T$ $\left(\boldsymbol{A}\right)$.

- **Dual Expansion Theorem in Terms of Eigenmatrices**

$$\boldsymbol{v} = \sum_{i=1}^{2n} a_i \boldsymbol{x}_i = \left[\boldsymbol{x}_1 \boldsymbol{x}_2 \ldots \boldsymbol{x}_{2n}\right] \begin{Bmatrix} a_1 \\ a_2 \\ . \\ a_{2n} \end{Bmatrix} = \boldsymbol{X}\boldsymbol{a} \Leftrightarrow \boldsymbol{v} = \sum_{i=1}^{2n} b_i \boldsymbol{y}_i = \left[\boldsymbol{y}_1 \boldsymbol{y}_2 \ldots \boldsymbol{y}_{2n}\right] \begin{Bmatrix} b_1 \\ b_2 \\ . \\ b_{2n} \end{Bmatrix} = \boldsymbol{Y}\boldsymbol{b}$$

$\downarrow \Leftarrow$ Premultiply by $\boldsymbol{Y}^T; \boldsymbol{Y}^T \boldsymbol{X} = \boldsymbol{I}$ $\qquad$ $\downarrow \Leftarrow$ Premultiply by $\boldsymbol{X}^T; \boldsymbol{X}^T \boldsymbol{Y} = \boldsymbol{I}$

$\boldsymbol{a} = \boldsymbol{Y}^T \boldsymbol{v}$ $\qquad$ $\boldsymbol{b} = \boldsymbol{X}^T \boldsymbol{v}$

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Returning to the Time Domain Physical Response!

$$\boldsymbol{x}(t) = \sum_{i=1}^{2n} a_i \boldsymbol{x}_i e^{\lambda_i t} \ : \ \text{Homogeneous solution}$$

$$\downarrow \Leftarrow \begin{cases} \text{Initial condition:} \boldsymbol{x}(0) = \sum_{i=1}^{2n} a_i \boldsymbol{x}_i \\ \qquad\qquad \downarrow \\ \qquad \boldsymbol{y}_j^T \boldsymbol{x}(0) = a_j, j = 1,2,\ldots,2n \end{cases}$$

$$= \sum_{i=1}^{2n} \underset{1\times 2n}{\boldsymbol{y}_i^T} \underset{2n\times 1}{\boldsymbol{x}}(0) \boldsymbol{x}_i e^{\lambda_i t} = \sum_{i=1}^{2n} \boldsymbol{x}_i e^{\lambda_i t} \boldsymbol{y}_i^T \boldsymbol{x}(0)$$

$$= \underset{2n\times 2n}{\boldsymbol{X}} \underset{2n\times 2n\text{: diagonal}}{e^{\Lambda t}} \underset{2n\times 2n}{\boldsymbol{Y}^T} \underset{2n\times 1}{\boldsymbol{x}(0)}$$

$\downarrow \Leftarrow \boldsymbol{x}(t) = e^{\boldsymbol{A}t} \boldsymbol{x}(0)$ for $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}(t)$ (See p.6, p.10, Lecture 3: Transition Matrix)

$\therefore \ e^{\boldsymbol{A}t} = \boldsymbol{X} e^{\Lambda t} \boldsymbol{Y}^T$: Computation of transition matrix in modal domain

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Another Proof of $e^{At} = Xe^{\Lambda t}Y^T$:

From Tayler Series Expansion :

$$e^{\Lambda t} = I + t\Lambda + \frac{t^2}{2!}\Lambda^2 + \frac{t^3}{3!}\Lambda^3 + \cdots$$

$$e^{At} \equiv I + tA + \frac{t^2}{2!}A^2 + \frac{t^3}{3!}A^3 + \cdots$$

From Biorthogonality:

$$Y^T X = I \rightarrow XY^T = I \quad (\because X^{-1} = Y^T) \ \& \ \ Y^T AX = \Lambda \rightarrow A = Y^{-T}\Lambda X^{-1} = X\Lambda Y^T$$

We have:

$$e^{At} = XY^T + tX\Lambda Y^T + \frac{t^2}{2!}X\Lambda \underbrace{Y^T X}_{I} \Lambda Y^T + \frac{t^3}{3!}X\Lambda \underbrace{Y^T X}_{I} \Lambda \underbrace{Y^T X}_{I} \Lambda Y^T + \cdots$$

$$= XY^T + tX\Lambda Y^T + \frac{t^2}{2!}X\Lambda^2 Y^T + \frac{t^3}{3!}X\Lambda^3 Y^T + \cdots$$

$$= X\left(I + t\Lambda + \frac{t^2}{2!}\Lambda^2 + \frac{t^3}{3!}\Lambda^3 + \cdots\right)Y^T = Xe^{\Lambda t}Y^T \quad \textbf{Q.E.D}$$

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• Similarity Transformation of a Square Matrix

$|\boldsymbol{A}-\lambda\boldsymbol{I}|=0$ for eigenvalues of $\boldsymbol{A} \Leftarrow [\boldsymbol{A}-\lambda\boldsymbol{I}]\boldsymbol{v}=0 \Leftarrow \boldsymbol{A}\boldsymbol{v}=\lambda\boldsymbol{v}$: Eigenvalue problem

$\downarrow \Leftarrow \boldsymbol{A}=\boldsymbol{P}\boldsymbol{B}\boldsymbol{P}^{-1}$: Similarity transformation of $\boldsymbol{B}$

$$|\boldsymbol{P}\boldsymbol{B}\boldsymbol{P}^{-1}-\lambda\boldsymbol{I}|=0$$

$$\downarrow \Leftarrow \boldsymbol{I}=\boldsymbol{P}\boldsymbol{I}\boldsymbol{P}^{-1}$$

$$|\boldsymbol{P}\boldsymbol{B}\boldsymbol{P}^{-1}-\lambda\boldsymbol{P}\boldsymbol{I}\boldsymbol{P}^{-1}|=0$$

$$\downarrow$$

$$|\boldsymbol{P}[\boldsymbol{B}-\lambda\boldsymbol{I}]\boldsymbol{P}^{-1}|=|\boldsymbol{P}||\boldsymbol{B}-\lambda\boldsymbol{I}||\boldsymbol{P}^{-1}|=0$$

$$\downarrow \Leftarrow |\boldsymbol{P}||\boldsymbol{P}^{-1}|=1 \Leftarrow \boldsymbol{P}\boldsymbol{P}^{-1}=\boldsymbol{I}$$

$$\therefore\ |\boldsymbol{B}-\lambda\boldsymbol{I}|=0 \quad \textbf{Q.E.D}$$

<u>Note:</u> Similarity transformation of a square matrix (use of transformed one-to-one mapping coordinates or states for the same vibration system) **does not change the eigenvalues**!! But **Eigenvectors will change.**

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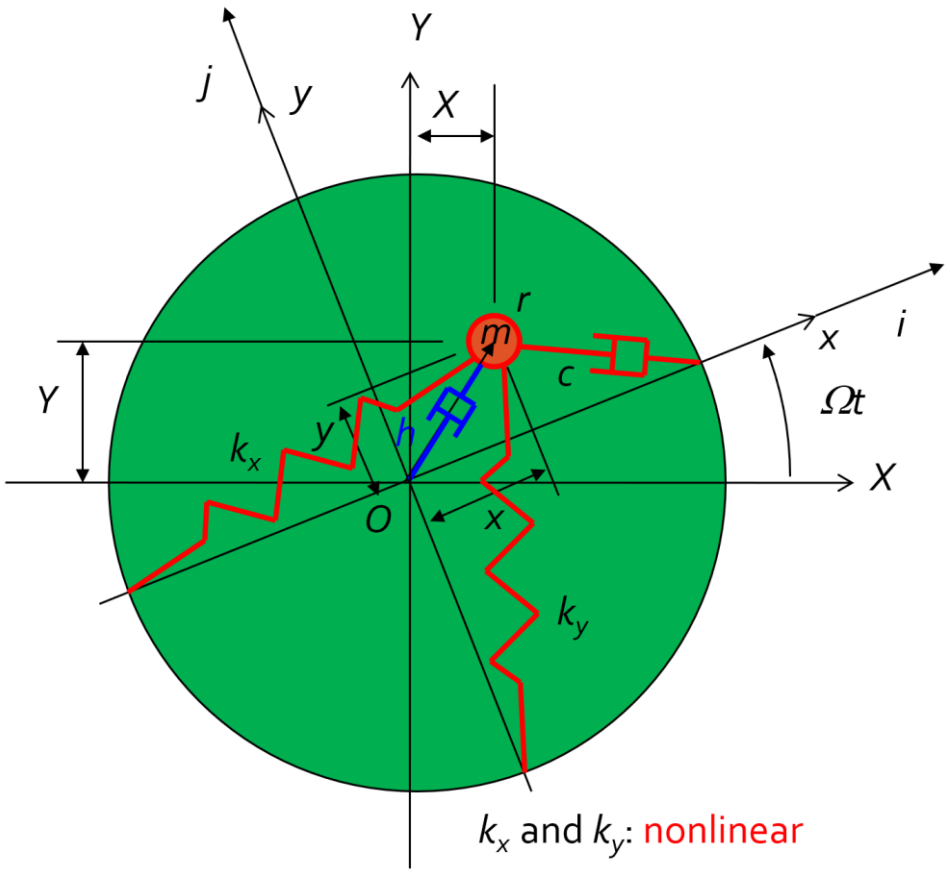
Ex.4.8: Linearized Mass-Damper-Spring System on Rotating Disk



$m = 1$ kg, $\Omega = 2$ rad/s, $k_x = 5$ N/m, $k_y = 10$ N/m

$c = 0.1$ Ns/m, $h = 0.2$ Ns/m

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \left(\boldsymbol{C}^* + \boldsymbol{G}\right)\dot{\boldsymbol{q}} + \left(\boldsymbol{K} + \boldsymbol{H}\right)\boldsymbol{q}(t) = 0$$

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \end{Bmatrix} + \begin{bmatrix} c+h & -2m\Omega \\ 2m\Omega & h \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{Bmatrix} + \begin{bmatrix} k_x - m\Omega^2 + 3k_x\varepsilon_x x_0^{\,2} & -h\Omega \\ h\Omega & k_y - m\Omega^2 + 3k_y\varepsilon_y y_0^{\,2} \end{bmatrix}\begin{Bmatrix} x_1 \\ y_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$\downarrow \Leftarrow E_1 : x_0 = y_0 = 0$

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \end{Bmatrix} + \begin{bmatrix} c+h & -2m\Omega \\ 2m\Omega & h \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{Bmatrix} + \begin{bmatrix} k_x - m\Omega^2 & -h\Omega \\ h\Omega & k_y - m\Omega^2 \end{bmatrix}\begin{Bmatrix} x_1 \\ y_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \text{ at } E_1$$

$\downarrow$

$$\underbrace{\frac{d}{dt}\begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix}}_{\dot{\boldsymbol{x}}} = \underbrace{\begin{bmatrix} 0 & \boldsymbol{I} \\ -\boldsymbol{M}^{-1}\left(\boldsymbol{K}+\boldsymbol{H}\right) & -\boldsymbol{M}^{-1}\left(\boldsymbol{C}^*+\boldsymbol{G}\right) \end{bmatrix}}_{\equiv \boldsymbol{A}:\ (2n\times 2n) \text{ real unsymmetric}} \underbrace{\begin{Bmatrix} \boldsymbol{q}(t) \\ \dot{\boldsymbol{q}}(t) \end{Bmatrix}}_{\equiv \boldsymbol{x}(t)} = \underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0.4 & -0.3 & 4 \\ -0.4 & -6 & -4 & -0.2 \end{bmatrix}}_{\equiv \boldsymbol{A}:\ (4\times 4) \text{ real unsymmetric}} \begin{Bmatrix} \boldsymbol{q}(t) \\ \dot{\boldsymbol{q}}(t) \end{Bmatrix}$$

$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}(t)$

$\downarrow \Leftarrow \boldsymbol{x}(t) = \boldsymbol{x}e^{\lambda t}, \lambda = \sigma + i\omega$ & $\boldsymbol{x}$: complex

$\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$: Eigenvalue problem

$\boldsymbol{y}^T\boldsymbol{A} = \lambda\boldsymbol{y}^T$: Adjoint eigenvalue problem

$\downarrow \Leftarrow$ Matlab: function [X,D]=eig(A) or [Y,D]=eig(A')

$\lambda_j, \boldsymbol{x}_j, \boldsymbol{y}_j, j = 1, 2, \ldots, 2n$

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Eigenvalues

```
>>  A=[0 0 1 0;
    0 0 0 1;
    -1 0.4 -0.3 4;
    -0.4 -6 -4 -0.2]
```

Solutions in Matlab

λ_3 = -0.1372 + 4.7653i $= \sigma_3 + i\omega_3$

λ_4 = -0.1372 - 4.7653i $= \sigma_4 + i\omega_4 = \sigma_3 - i\omega_3$

λ_1 = -0.1128 + 0.5083i $= \sigma_1 + i\omega_1$

λ_2 = -0.1128 - 0.5083i $= \sigma_2 + i\omega_2 = \sigma_1 - i\omega_1$

Solutions in text

Eigenvalues are indexing from the lowest!(different indexing from the matlab output):

Eigenvectors

```
>>  [X,D]=eig(A)
X =
   0.1354 - 0.0021i   0.1354 + 0.0021i   0.8359 + 0.0000i   0.8359 + 0.0000i
   0.0044 + 0.1543i   0.0044 - 0.1543i   0.0081 - 0.2965i   0.0081 + 0.2965i
  -0.0087 + 0.6453i  -0.0087 - 0.6453i  -0.0943 + 0.4249i  -0.0943 - 0.4249i
  -0.7358 + 0.0000i  -0.7358 + 0.0000i   0.1498 + 0.0376i   0.1498 - 0.0376i

D =
  -0.1372 + 4.7653i   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i
   0.0000 + 0.0000i  -0.1372 - 4.7653i   0.0000 + 0.0000i   0.0000 + 0.0000i
   0.0000 + 0.0000i   0.0000 + 0.0000i  -0.1128 + 0.5083i   0.0000 + 0.0000i
   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i  -0.1128 - 0.5083i

>> [Y,D]=eig(A')
Y =
   0.0383 + 0.1022i   0.0383 - 0.1022i  -0.0289 - 0.4078i  -0.0289 + 0.4078i
   0.6841 + 0.0000i   0.6841 + 0.0000i   0.8846 + 0.0000i   0.8846 + 0.0000i
   0.4732 + 0.0477i   0.4732 - 0.0477i  -0.2116 - 0.0013i  -0.2116 + 0.0013i
   0.0472 - 0.5401i   0.0472 + 0.5401i   0.0025 - 0.0750i   0.0025 + 0.0750i

D =
  -0.1372 + 4.7653i   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i
   0.0000 + 0.0000i  -0.1372 - 4.7653i   0.0000 + 0.0000i   0.0000 + 0.0000i
   0.0000 + 0.0000i   0.0000 + 0.0000i  -0.1128 + 0.5083i   0.0000 + 0.0000i
   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i  -0.1128 - 0.5083i
```


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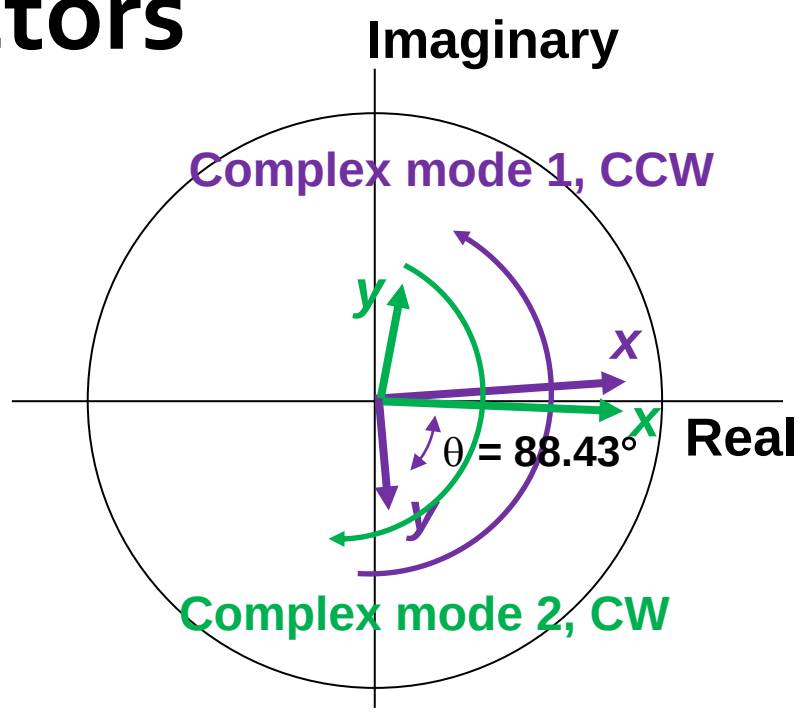
Representation of mode shape vectors in polar plot for complex numbers



Eigenvalues(Eq. (c) on p.195, re-indexing in ascending order):

λ_1 = **-0.1128 + 0.5083i** $= \sigma_1 + i\omega_1$

λ_2 = **-0.1128 - 0.5083i** $= \sigma_1 - i\omega_1$

λ_3 = -0.1372 + 4.7653i

λ_4 = -0.1372 - 4.7653i

Eigenmatrix X (Eq. (d) on p. 195 re-indexing in ascending order) =

[**0.8359 + 0.0000i**	**0.8359 + 0.0000i**	0.1354 - 0.0021i	0.1354 + 0.0021i
0.0081 - 0.2965i	**0.0081 + 0.2965i**	0.0044 + 0.1543i	0.0044 - 0.1543i
-0.0943 + 0.4249i	-0.0943 - 0.4249i	-0.0087 + 0.6453i	-0.0087 - 0.6453i
0.1498 + 0.0376i	0.1498 - 0.0376i	-0.7358 + 0.0000i	-0.7358 + 0.0000i]

```
>> theta1=(atan2(0.0000,0.8359)…
    -atan2(- 0.2965,0.0081))*180/pi

theta1 = 88.4351
```

Note:

(x, y): first mode shape vector rotating **counterclockwise** $\boldsymbol{x}_1 e^{\lambda_1 t}, \lambda_1 = \sigma_1 + i\omega_1$

(x, y): conjugate vector rotating **clockwise** $\boldsymbol{x}_2 e^{\lambda_2 t}, \lambda_2 = \sigma_1 - i\omega_1$ **to have a real valued response !**

Angle θ : Phase difference between two adjacent responses in x and y in a specific mode

θ (=1.54rad)/ ω_1(0.5083 rad/s): Time difference between two adjacent peaks in x and y

Time varying coordinates ($x(t)$, $y(t)$) are determined by the initial conditions.

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Representation of motion at the first mode in the *x-y* coordinate

Eigenvalues(Eq. (c) on p.195)→

λ_1 = **-0.1128 + 0.5083i**(+: counterclockwise)

λ_2 = **-0.1128 - 0.5083i**(-: clockwise)

λ_3 = -0.1372 + 4.7653i

λ_4 = -0.1372 - 4.7653i

Eigenmatrix X (Eq. (d) on p. 195) =

[**0.8359 + 0.0000i**	**0.8359 + 0.0000i**	0.1354 - 0.0021i	0.1354 + 0.0021i
0.0081 - 0.2965i	**0.0081 + 0.2965i**	0.0044 + 0.1543i	0.0044 - 0.1543i
-0.0943 + 0.4249i	-0.0943 - 0.4249i	-0.0087 + 0.6453i	-0.0087 - 0.6453i
0.1498 + 0.0376i	0.1498 - 0.0376i	-0.7358 + 0.0000i	-0.7358 + 0.0000i]

↓⇐ Proper initial conditions: $x(0), y(0), \dot{x}(0), \dot{y}(0)$

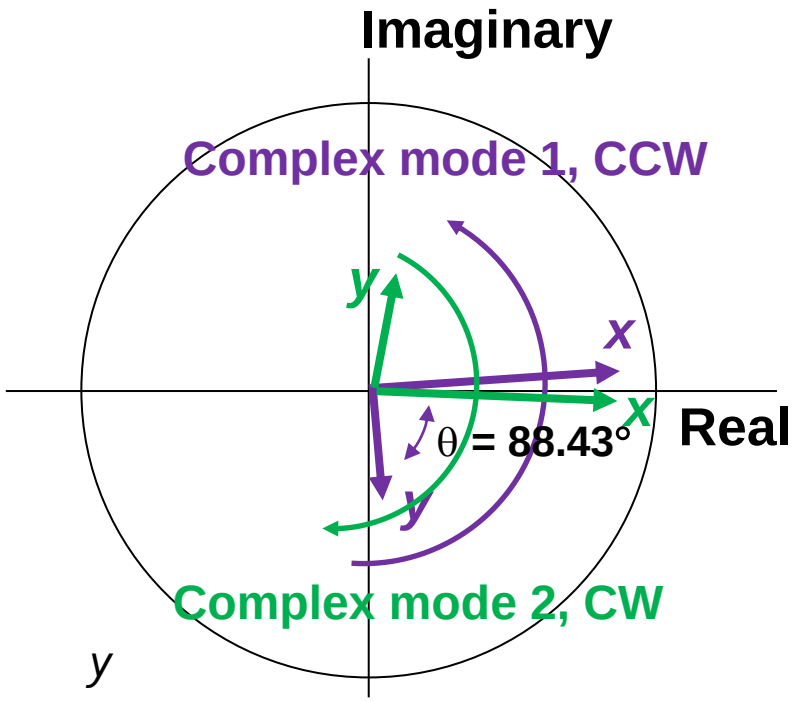
$$\begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} = c \begin{Bmatrix} 0.8359 + 0.0000i? \\ 0.0081 - 0.2965i? \end{Bmatrix} e^{(-0.1128+0.5083i)t} + c^* \begin{Bmatrix} 0.8359 + 0.0000i? \\ 0.0081 + 0.2965i? \end{Bmatrix} e^{(-0.1128-0.5083i)t}$$

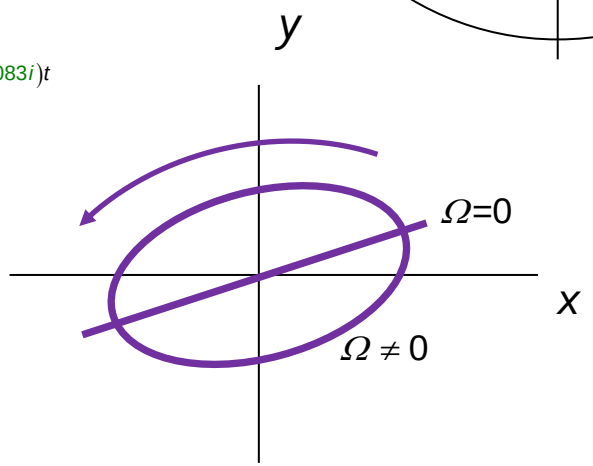
$$= \text{real constant} \times \begin{Bmatrix} 0.8359 \\ 0.2966 \end{Bmatrix} e^{-0.1128t} \begin{Bmatrix} \cos 0.5083t \\ \cos(0.5083t - 1.5433) \end{Bmatrix}$$

↓

Trajectory on the disk: Ellipses with a slope (a line for Ω=0) and decreasing majors and minors ⇒

View in fixed frame: The above ellipses rotating at the disk speed





First Mode: Forward Whirl Motion on Ellipse!

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Summary of 4.6, 4.7 and 4.8

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \left(\underset{\text{sym}}{\boldsymbol{C}^*} + \underset{\text{skew sym}}{\boldsymbol{G}}\right)\dot{\boldsymbol{q}} + \left(\underset{\text{sym}}{\boldsymbol{K}} + \underset{\text{skew sym}}{\boldsymbol{H}}\right)\boldsymbol{q}(t) = 0$$

Natural system with proportional damping:

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = 0, \boldsymbol{C} = \alpha\boldsymbol{M} + \beta\boldsymbol{K}$$

$$\downarrow\Leftarrow \begin{cases} \boldsymbol{K}\boldsymbol{u} = \lambda\boldsymbol{M}\boldsymbol{u}, \lambda = -s^2 = \omega^2 \\ \downarrow\Leftarrow \boldsymbol{u} = \boldsymbol{Q}^{-1}\boldsymbol{v}, \text{ where } \boldsymbol{M} = \boldsymbol{Q}^T\boldsymbol{Q} \\ \boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v} \end{cases}$$

$$\boldsymbol{q}(t) = \sum_{r=1}^{n} \boldsymbol{u}_r \left(b_r e^{\left(-\varsigma_r\omega_r + i\sqrt{1-\varsigma_r^2}\omega_r\right)t} + \bar{b}_r e^{\left(-\varsigma_r\omega_r - i\sqrt{1-\varsigma_r^2}\omega_r\right)t} \right)$$

$$= \sum_{r=1}^{n} a_r \boldsymbol{u}_r e^{-\varsigma_r\omega_r t} \cos\left(\sqrt{1-\varsigma_r^2}\,\omega_r t + \phi_r\right)$$

Gyroscopic conservative system:

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{G}\dot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = 0$$

$$\downarrow\Leftarrow \begin{cases} \text{Augmentation by } \boldsymbol{K}\dot{\boldsymbol{q}} = \boldsymbol{K}\dot{\boldsymbol{q}} \\ \underbrace{\begin{bmatrix} \boldsymbol{K} & 0 \\ 0 & \boldsymbol{M} \end{bmatrix}}_{\equiv \boldsymbol{M}^*:2n\times 2n \text{ sym}} \underbrace{\frac{d}{dt}\begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix}}_{\dot{\boldsymbol{x}}} = -\underbrace{\begin{bmatrix} 0 & -\boldsymbol{K} \\ \boldsymbol{K} & \boldsymbol{G} \end{bmatrix}}_{\equiv \boldsymbol{G}^*:\ 2n\times 2n \text{ skew sym}} \underbrace{\begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix}}_{\boldsymbol{x}} \\ \downarrow\Leftarrow \begin{cases} i\omega\boldsymbol{M}^*\boldsymbol{x} = -\boldsymbol{G}^*\boldsymbol{x}, \boldsymbol{x} = \boldsymbol{y} + i\boldsymbol{z} \\ \lambda\boldsymbol{M}^*\boldsymbol{y} = \boldsymbol{K}^*\boldsymbol{y}, \boldsymbol{K}^* \equiv \boldsymbol{G}^{*T}\boldsymbol{M}^{*-1}\boldsymbol{G}^* \\ \boldsymbol{y} = \boldsymbol{Q}^{*-1}\boldsymbol{v}, \text{ where } \boldsymbol{M}^* = \boldsymbol{Q}^{*T}\boldsymbol{Q}^* \end{cases} \\ \lambda\boldsymbol{v} = \underset{2n\times 2n}{\boldsymbol{A}}\ \underset{2n\times 1}{\boldsymbol{v}},\ \boldsymbol{A} = \boldsymbol{Q}^{*-T}\boldsymbol{K}^*\boldsymbol{Q}^{*-1} \end{cases}$$

$$\begin{Bmatrix} \boldsymbol{q}(t) \\ \dot{\boldsymbol{q}}(t) \end{Bmatrix} = \sum_{r=1}^{n}\left(\boldsymbol{x}_r e^{i\omega_r t} + \bar{\boldsymbol{x}}_r e^{-i\omega_r t}\right), \text{ where}$$

$$\underset{2n\times 1}{\boldsymbol{x}_r} = \boldsymbol{y}_r + i\boldsymbol{z}_r = \boldsymbol{Q}^{*-1}\left(\boldsymbol{v}_{2r-1} + i\boldsymbol{v}_{2r}\right) = \begin{Bmatrix} \underset{n\times 1}{\boldsymbol{u}_r} \\ \underset{n\times 1}{i\omega_r\boldsymbol{u}_r} \end{Bmatrix}$$

General system:

$$\ddot{\boldsymbol{q}} = -\boldsymbol{M}^{-1}\left(\boldsymbol{C}^* + \boldsymbol{G}\right)\dot{\boldsymbol{q}} - \boldsymbol{M}^{-1}\left(\boldsymbol{K} + \boldsymbol{H}\right)\boldsymbol{q}$$

$$\downarrow\Leftarrow \text{Augmentation by } \dot{\boldsymbol{q}} = \dot{\boldsymbol{q}}$$

$$\underbrace{\frac{d}{dt}\begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix}}_{\dot{\boldsymbol{x}}} = \underbrace{\begin{bmatrix} 0 & \boldsymbol{I} \\ -\boldsymbol{M}^{-1}\left(\boldsymbol{K} + \boldsymbol{H}\right) & -\boldsymbol{M}^{-1}\left(\boldsymbol{C}^* + \boldsymbol{G}\right) \end{bmatrix}}_{\equiv \boldsymbol{A}:\ (2n\times 2n) \text{ unsymmetric matrix}} \underbrace{\begin{Bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{Bmatrix}}_{\boldsymbol{x}:\ \text{state vector}} :$$

$$\downarrow\Leftarrow \begin{cases} \dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}(t) \\ \boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}, \boldsymbol{A}^T\boldsymbol{y} = \lambda\boldsymbol{y} \\ \boldsymbol{y}_i^T\boldsymbol{x}_j(0) = \delta_{ij} \end{cases}$$

$$\begin{Bmatrix} \boldsymbol{q}(t) \\ \dot{\boldsymbol{q}}(t) \end{Bmatrix} = \sum_{i=1}^{2n} a_i\boldsymbol{x}_i e^{\lambda_i t} = \sum_{i=1}^{n}\left(a_i\boldsymbol{x}_i e^{\lambda_i t} + \bar{a}_i\bar{\boldsymbol{x}}_i e^{\bar{\lambda}_i t}\right)$$

$$= \sum_{i=1}^{2n} \underset{1\times 2n}{\boldsymbol{y}_i^T}\ \underset{2n\times 1}{\boldsymbol{x}}(0)\boldsymbol{x}_i e^{\lambda_i t} = \sum_{i=1}^{2n}\boldsymbol{x}_i e^{\lambda_i t}\boldsymbol{y}_i^T\boldsymbol{x}(0)$$

$$= \underset{2n\times 2n}{\boldsymbol{X}}\ \underset{2n\times 2n:\ \text{diagonal}}{e^{\boldsymbol{\Lambda}t}}\ \underset{2n\times 2n}{\boldsymbol{Y}^T}\ \underset{2n\times 1}{\boldsymbol{x}(0)}$$

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Highlight 13

- The general case of vibration including mass, stiffness, damping, and gyroscopic terms in equation of motion lead us to have unsymmetric eigenvalue problem to have complex-valued eigenvalues and eigenvectors which are complex conjugate pairs.

- The unsymmetric eigenvalue problem can be constructed by state-space representation and its adjoint eigenvalue problem give us bi-orthogonality between right and left eigenvectors.

- A complex modeshape represents phase difference between degree of freedoms, which is in general arbitrary value: 0 and 180 degree for conservative natural system, 90 degree for gyroscopic conservative system, arbitrary value for general nonconservative systems.

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